Equilibrium Compensation Based Control: A Universal Control Scheme for Systems With Mismatched Disturbances and Sensor Errors

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***Abstract*—There are significant couplings between control techniques and disturbance rejection techniques in traditional control schemes for systems with mismatched disturbances, such that both techniques have to be designed simultaneously. In this paper, independent of specific control methods, a universal disturbance rejection scheme named equilibrium compensation based control, is proposed for general systems to attenuate mismatched disturbances and sensor errors while retaining the nominal controller's performance. In addition, several criteria are provided to analyze the observability and compensability of the disturbances and sensor errors. Simulations and experiments of tracking tasks for a single-track-two-wheeled robot with four different controllers demonstrate the effectiveness of the proposed scheme.**

***Index Terms*—Disturbance rejection, equilibrium compensation based control (ECBC), mismatched disturbances, sensor errors, single-track-two-wheeled (STTW) robot.**

I. Introduction

DISTURBANCES broadly exist in various physical control systems, such as single-track-two-wheeled (STTW) robots [1], [2], [3], underwater robots [4], [5], and power converters [6], [7], which bring significant challenges to controller design. Especially, constant and slow-varying disturbances always yield steady-state errors, which may degrade tracking control performances. To achieve accurate tracking, a robust controller is desired to compensate the effects of disturbance from the controlled output. Two fundamental approaches to reject disturbances are disturbance observer based control (DOBC) [7], [8], [9], [10], [11], [12] and integrator based control [5], [13], [14]. The first one gains more preference in the literature because of the following reasons: 1) DOBC is flexible, since the original problem is divided into two subproblems: nominal controller design and disturbance attenuation [15], [16]; 2) it is well-known that integral action may bring overshoot and long settling time [8].

Different from matched disturbances that can be directly compensated by control inputs, mismatched disturbances are more difficult to attenuate and thus draw more attentions [7], [8], [9], [10], [11], [12], [17], [18], [19], [20], [21]. In the field of active disturbance rejection control (ADRC) [22], extended state observer (ESO) can only address essential chain of integrators with matched disturbances. To get rid of this restriction, the generalized extended state observer based control (GESOBC) was proposed in [10], which can be applied to more general systems subject to mismatched disturbances [11], [12]. Even though the disturbance observer in GESOBC is not limited to GESO, the controller is coupled with disturbance rejection, which cannot be directly migrated to other control methods. Besides, the existence condition of GESOBC depends on the control gains, while this article reveals that the condition is equivalent to a concise form only depending on the open-loop information.

In research of model predictive control (MPC), to address model-plant mismatch, the offset-free MPC [17], [18], [19], [20] was developed, which is based on the internal model principle. The key concept of these methods is to augment the system model with a disturbance model such that the mismatch between the measured and predicted outputs in steady state can be obtained by an observer. Then, the estimated disturbances are brought into the initialization of the optimization problem of MPC, which attains the offset-free control. However, the controlled outputs in offset-free MPC are linear combinations of measured outputs that still have disturbances, such that the offset-free tracking is with respect to the noised outputs.

Studies on sliding mode control (SMC) have incorporated disturbance observer (DOB) to address mismatched disturbances [6], [7], [8], [9]. By designing a novel sliding mode surface containing estimated disturbances, these methods can attenuate mismatched disturbances while retaining nominal performance. The restriction of these methods is that the plants are limited to chain of integrators, such that they cannot handle nonminimum phase systems with unstable internal dynamics. Even though some methods such as dynamic sliding-mode control (DSMC) [23] are proposed for nonminimum phase systems,

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there is rarely research considering nonminimum phase property and mismatched disturbances simultaneously.

Various control fields have tackled the issue of mismatched disturbance rejection using method-specific languages, but these methods cannot be directly migrated to general controllers. To address this, the equilibrium compensation based control (ECBC) is proposed in this paper, which can be used as an extension to existing control methods and is applicable to general systems, including nonminimum phase systems with mismatched disturbances.

The ECBC scheme is based on the two ideas: that the steady-state output of a system only depends on the equilibrium to which it converges, and that the objective of tracking control is to force the system to a specific equilibrium without steady-state output error. Concretely, the ECBC scheme consists of three main steps. First, a preferred control method for the nominal (i.e., disturbance-free) system and a DOB are selected, ensuring that both the nominal closed-loop system and the estimation error system are input-to-state stable (ISS). Second, an equilibrium compensator (EC) is designed to obtain an appropriate equilibrium based on the estimated disturbances, the equilibrium equation, and the controlled output equation. Finally, the controller, the observer, and the EC are combined to convert the perturbed system to its nominal form. Consequently, the ECBC scheme can completely remove the effects of mismatched disturbances from the steady-state of the controlled output, as long as certain conditions are met.

The main contributions of this paper include: 1) proposing the ECBC scheme, which decouples controllers and disturbance rejection techniques such that numerous existing control methods can be directly extended to systems with mismatched disturbances; 2) providing several new criteria for analyzing the observability and compensability of the disturbances and sensor errors; 3) proposing the output decomposition method and the minimum-norm compensation method for multiple-input multiple-output (MIMO) systems; 4) verifying the proposed scheme on an STTW robot by a state feedback controller and an SMC controller.

This rest of this paper is organized as follows. In Section II, ECBC for single-input single-output (SISO) systems is introduced, and its fundamental properties are claimed. In Section III, ECBC is extended to MIMO systems, and additional compensability criteria and variant methods are provided. Section IV discusses sensor errors and provides observability and compensability criteria for such errors. In Section V, the simulation and experimental results of an STTW robot are presented. Finally, Section VI concludes this paper.

II. ECBC for SISO Systems

A. System Formulation

In this section, we focus on the following linear SISO system with a single disturbance:

$$\begin{aligned}\dot{\boldsymbol{x}} &= \boldsymbol{A}\boldsymbol{x} + \boldsymbol{b}_u u + \boldsymbol{b}_d d\\ \boldsymbol{y}_m &= \boldsymbol{C}_m \boldsymbol{x}\\ y_o &= \boldsymbol{c}_o \boldsymbol{x}\end{aligned} \tag{1}$$

where $\boldsymbol{x} \in \mathbb{R}^n$, $u \in \mathbb{R}$, $d \in \mathbb{R}$, $\boldsymbol{y}_m \in \mathbb{R}^{n_m}$, and $y_o \in \mathbb{R}$ are the state, input, disturbance, measurable output, and controlled output, and $\boldsymbol{A}$, $\boldsymbol{b}_u$, $\boldsymbol{b}_d$, $\boldsymbol{C}_m$, $\boldsymbol{c}_o$ have appropriate dimensions. The lumped disturbance d consists of parameter perturbations, unmodeled dynamics, and external disturbances.

To facilitate the following definition of the mismatched disturbance, we introduce the input matrix $\boldsymbol{B}_u$ corresponding to MIMO systems.

Definition 1 (see[10], [15], [16], [24], [25]): The disturbance in (1) is defined as a matched disturbance if $\boldsymbol{b}_d \in$ Range $(\boldsymbol{B}_u)$, i.e., $\text{rank}(\boldsymbol{B}_u) = \text{rank}([\boldsymbol{B}_u \quad \boldsymbol{b}_d])$. Otherwise, it is called a mismatched disturbance.

Assumption 1: The disturbance d is differentiable and bounded, and has a steady value, i.e., $\lim_{t\to\infty} d = d^*$. The derivative $h = \dot{d}$ is bounded and $\lim_{t\to\infty} h = 0$.

Assumption 1 is reasonable since the common disturbances in practical systems include: constant parameter perturbations of stable systems, invariant unmodeled dynamics of stable systems and slow-varying external disturbances, all of which satisfy Assumption 1.

To design a DOB, the extended state is defined as $\bar{\boldsymbol{x}} = [\boldsymbol{x}^T, d]^T$. The extended system is written as

$$\begin{aligned}\dot{\bar{\boldsymbol{x}}} &= \bar{\boldsymbol{A}}\bar{\boldsymbol{x}} + \bar{\boldsymbol{b}}_u u + \bar{\boldsymbol{b}}_d h\\ \boldsymbol{y}_m &= \bar{\boldsymbol{C}}_m \bar{\boldsymbol{x}}\end{aligned} \tag{2}$$

where $h = \dot{d}$ and

$$\bar{\boldsymbol{A}} = \begin{bmatrix}\boldsymbol{A} & \boldsymbol{b}_d\\ \boldsymbol{0} & 0\end{bmatrix}, \bar{\boldsymbol{b}}_u = \begin{bmatrix}\boldsymbol{b}_u\\ 0\end{bmatrix}, \bar{\boldsymbol{b}}_d = \begin{bmatrix}\boldsymbol{0}\\ 1\end{bmatrix}, \bar{\boldsymbol{C}}_m = \begin{bmatrix}\boldsymbol{C}_m & \boldsymbol{0}\end{bmatrix}.$$

B. Observability of Disturbances

Assumption 2: $(\boldsymbol{A}, \boldsymbol{b}_u)$ is controllable and $(\boldsymbol{A}, \boldsymbol{C}_m)$ is observable.

Lemma 1: System $(\bar{\boldsymbol{A}}, \bar{\boldsymbol{C}}_m)$ is observable, if and only if system $(\boldsymbol{A}, \boldsymbol{C}_m)$ is observable and the matrix

$$\boldsymbol{Q}_{dm} = \begin{bmatrix}\boldsymbol{A} & \boldsymbol{b}_d\\ \boldsymbol{C}_m & \boldsymbol{0}\end{bmatrix}$$

has full column rank.

Proof: Let $\mathcal{O}$ denote the observability matrix of $(\boldsymbol{A}, \boldsymbol{C}_m)$. The observability matrix of $(\bar{\boldsymbol{A}}, \bar{\boldsymbol{C}}_m)$ can be decomposed as

$$\begin{aligned}\bar{\mathcal{O}} &= \begin{bmatrix}\bar{\boldsymbol{C}}_m\\ \bar{\boldsymbol{C}}_m\bar{\boldsymbol{A}}\\ \vdots\\ \bar{\boldsymbol{C}}_m\bar{\boldsymbol{A}}^n\end{bmatrix} = \underbrace{\begin{bmatrix}\boldsymbol{C}_m & \boldsymbol{0}_{n_m\times 1}\\ \boldsymbol{C}_m\boldsymbol{A} & \boldsymbol{C}_m\boldsymbol{b}_d\\ \vdots & \vdots\\ \boldsymbol{C}_m\boldsymbol{A}^n & \boldsymbol{C}_m\boldsymbol{A}^{n-1}\boldsymbol{b}_d\end{bmatrix}}_{\bar{\mathcal{O}}_A \quad \bar{\mathcal{O}}_B}\\ &= \underbrace{\begin{bmatrix}\boldsymbol{0} & \boldsymbol{I}_{n_m}\\ \mathcal{O} & \boldsymbol{0}\end{bmatrix}}_{\boldsymbol{P}_o}\underbrace{\begin{bmatrix}\boldsymbol{A} & \boldsymbol{b}_d\\ \boldsymbol{C}_m & \boldsymbol{0}\end{bmatrix}}_{\boldsymbol{Q}_{dm}}.\end{aligned} \tag{3}$$

Necessity: Assume $(\boldsymbol{A}, \boldsymbol{C}_m)$ is not observable, then $rank(\mathcal{O}) < n$. From Cayley–Hamilton theorem, $rank(\bar{\mathcal{O}}_A) = rank(\mathcal{O}) < n$. Hence, $rank(\bar{\mathcal{O}}) \le rank(\bar{\mathcal{O}}_A) + rank(\bar{\mathcal{O}}_B) < n + 1$,

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which implies that $(\bar{\boldsymbol{A}}, \bar{\boldsymbol{C}}_m)$ is not observable. Meanwhile, assume $\boldsymbol{Q}_{dm}$ is not of full column rank, then $rank(\bar{\mathcal{O}}) \leq rank(\boldsymbol{Q}_{dm}) < n+1$, $(\bar{\boldsymbol{A}}, \bar{\boldsymbol{C}}_m)$ is not observable.

Sufficiency: As $(\boldsymbol{A}, \boldsymbol{C}_m)$ is observable, $\boldsymbol{P}_o$ has full column rank, hence $rank(\bar{\mathcal{O}}) = rank(\boldsymbol{Q}_{dm}) = n+1$, $(\bar{\boldsymbol{A}}, \bar{\boldsymbol{C}}_m)$ is observable. ■

Remark 1: In traditional ESO related studies (e.g., [22]) and SMC-DO related studies (e.g., [8], [9]), the plants are limited to chain of integrators, such that the observability condition for general systems was not considered. In previous works of GESO (e.g., [10], [11], [12]), only one necessary condition that $(\boldsymbol{A}, \boldsymbol{C})$ should be observable was proposed, while our work completes the necessary and sufficient condition. In the literature of unknown input observer (UIO) [26], an alternative form named strong detectability condition was put forward. In the research of offset-free MPC [17], [27], a similar condition was proposed for discrete systems and proved by Hautus observability condition.

C. Observer and the Controller in ECBC

Before introducing the ECBC scheme, several assumptions for the general observers and controllers are stated.

In ECBC, we assume the observer has the general form

$$\dot{\hat{\tilde{\boldsymbol{x}}}} = \boldsymbol{\zeta}(\hat{\tilde{\boldsymbol{x}}}, u, \boldsymbol{y}_m). \tag{4}$$

Subtracting (2) by (4), the error system is denoted by

$$\dot{\bar{\boldsymbol{e}}} = \boldsymbol{\eta}(\bar{\boldsymbol{e}}, h) \tag{5}$$

where $\boldsymbol{\eta}$ is a general function and

$$\bar{\boldsymbol{e}} \equiv [\boldsymbol{e}_x^T, e_d]^T \equiv [(\boldsymbol{x} - \hat{\boldsymbol{x}})^T, d - \hat{d}]^T. \tag{6}$$

Assumption 3: The DOB (4) is designed such that the error system (5) is ISS. The function $\boldsymbol{\eta} : \mathbb{R}^{n+2} \to \mathbb{R}^{n+1}$ is a bounded mapping satisfying $\boldsymbol{\eta}(0,0) \to \boldsymbol{0}$ as $\bar{\boldsymbol{e}} \to \boldsymbol{0}, h \to 0$.

Remark 2: Assumption 3 implies that if h is bounded and $\lim_{t\to\infty} h = 0$ then the observation error $\bar{\boldsymbol{e}}$ is bounded and $\lim_{t\to\infty} \bar{\boldsymbol{e}} = \boldsymbol{0}$, i.e., $\lim_{t\to\infty} \hat{\boldsymbol{x}} = \boldsymbol{x}, \lim_{t\to\infty} \hat{d} = d$. Assumption 3 is a broadly applicable assumption that is already satisfied by many common observers, such as GESO [10], the observer for MPC [17] and the nonlinear DOB (NDOB) for SMC [8]. One sufficient condition for ISS is that $\dot{\bar{\boldsymbol{e}}} = \boldsymbol{\eta}(\bar{\boldsymbol{e}}, 0)$ has a globally exponentially stable equilibrium point at the origin and $\boldsymbol{\eta}(\bar{\boldsymbol{e}}, h)$ is globally Lipschitz [28, Lemma 4.5].

In ECBC, we assume the controller has the general form

$$u = \kappa(\boldsymbol{x}) \tag{7}$$

then the closed-loop nominal system is

$$\dot{\boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{b}_u \kappa(\boldsymbol{x}) \tag{8}$$

and the closed-loop nominal system with perturbations $\boldsymbol{\varepsilon}$ is

$$\dot{\boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{b}_u \kappa(\boldsymbol{x}) + \boldsymbol{\varepsilon} \equiv \boldsymbol{\xi}(\boldsymbol{x}, \boldsymbol{\varepsilon}). \tag{9}$$

Assumption 4: The controller (7) is designed such that the closed-loop system (9) is ISS.

Remark 3: Assumption 4 implies that if $\boldsymbol{\varepsilon}$ is bounded and $\lim_{t\to\infty} \boldsymbol{\varepsilon} = 0$ then the state $\boldsymbol{x}$ is bounded and $\lim_{t\to\infty} \boldsymbol{x} = \boldsymbol{0}$. Assumption 4 is an essential characteristic that is inherent to commonly utilized controllers, such as state feedback controller [10], MPC [17], and SMC [8].

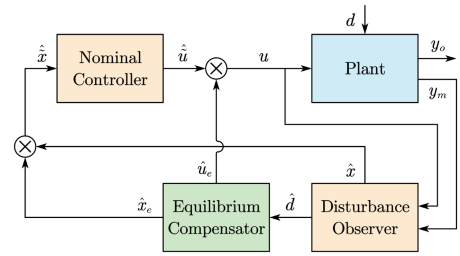

Fig. 1. Block diagram of the proposed ECBC scheme.

D. Equilibrium Compensator

Lemma 2: Consider system (1). If $(\boldsymbol{A}, \boldsymbol{b}_u)$ is controllable, then there exist infinitely many equilibriums $(\boldsymbol{x}_e, u_e)$, defined as the points satisfying the equilibrium equation

$$\boldsymbol{A}\boldsymbol{x}_e + \boldsymbol{b}_u u_e + \boldsymbol{b}_d d = \boldsymbol{0}. \tag{10}$$

Proof: As $(\boldsymbol{A}, \boldsymbol{b}_u)$ is controllable, based on Hautus lemma (also known as PBH test [29]), $[\lambda \boldsymbol{I} - \boldsymbol{A}, \boldsymbol{b}_u]$ has full row rank for all $\lambda \in \mathbb{C}$. We can obtain that matrix $[\boldsymbol{A} \quad \boldsymbol{b}_u]$ has full row rank, hence (10) has infinitely many solutions. ■

Remark 4: Lemma 2 implies that a controllable system does not have a unique equilibrium. If some disturbances are injected to the system, the equilibrium that the system converged to may not be the origin, resulting that the steady value of y_o is not equal to zero. For instance, consider a second-order system with a mismatched disturbance

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u + \begin{bmatrix} 1 \\ 0 \end{bmatrix} d$$

and $y = x_1$. The controller $u = -2x_1 - 3x_2$ stabilizes the disturbance-free system since the closed-loop poles are $[-1, -2]$. When $d = 1$, the state $\boldsymbol{x}$ and input u will converge to $(x_1, x_2, u) = (1.5, -1, 0)$ with $y = 1.5$. However, there still exists an equilibrium $(x_1, x_2, u) = (0, -1, 0)$ such that $y = 0$, which offers the opportunity to eliminate the effect of disturbances from the controlled output.

Inspired by Remark 4, the ECBC scheme, as depicted in Fig. 1, is designed to drive the state to a desired equilibrium, and its design procedures are introduced as follows.

1) Select a desired equilibrium $(\hat{\boldsymbol{x}}_e, \hat{u}_e)$ satisfying the equilibrium equation

$$\boldsymbol{A}\hat{\boldsymbol{x}}_e + \boldsymbol{b}_u \hat{u}_e + \boldsymbol{b}_d \hat{d} = \boldsymbol{0} \tag{11}$$

where $\hat{d}$ is obtained from observer (4). Then, $[\hat{\boldsymbol{x}}_e^T, \hat{u}_e^T]^T$ can be formulated as a function of $\hat{d}$, such that $[\hat{\boldsymbol{x}}_e^T, \hat{u}_e^T]^T = [\boldsymbol{\sigma}_x^T(\hat{d}), \sigma_u^T(\hat{d})]^T \equiv \boldsymbol{\sigma}(\hat{d})$. According to Lemma 2, the solution is not unique and thus the function $\boldsymbol{\sigma}(\cdot)$ depends on specific task. Section II-E will give the expression of $\boldsymbol{\sigma}(\cdot)$ for accurate tracking.

2) Construct the compensated state and control input as

$$\hat{\tilde{\boldsymbol{x}}} = \hat{\boldsymbol{x}} - \hat{\boldsymbol{x}}_e, \qquad \hat{\tilde{u}} = u - \hat{u}_e \tag{12}$$

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where $\hat{\boldsymbol{x}}$ is obtained from observer (4).

3) Design the controller as

$$u = \kappa(\hat{\tilde{\boldsymbol{x}}}) + \hat{u}_e. \tag{13}$$

Combining (6) and (12) gives

$$\hat{\tilde{\boldsymbol{x}}} = \boldsymbol{x} - \hat{\boldsymbol{x}}_e - \boldsymbol{e}_x. \tag{14}$$

By taking the derivative of $\hat{\tilde{\boldsymbol{x}}}$ and using (1), (5), and (11), we have

$$\begin{aligned}\dot{\hat{\tilde{\boldsymbol{x}}}} &= \dot{\boldsymbol{x}} - \dot{\hat{\boldsymbol{x}}}_e - \dot{\boldsymbol{e}}_x - (\boldsymbol{A}\hat{\boldsymbol{x}}_e + \boldsymbol{b}_u\hat{u}_e + \boldsymbol{b}_d\hat{d}) \\ &= \boldsymbol{A}(\boldsymbol{x} - \hat{\boldsymbol{x}}_e) + \boldsymbol{b}_u(u - \hat{u}_e) + \boldsymbol{b}_d(d - \hat{d}) - \boldsymbol{\eta}_x(\bar{\boldsymbol{e}}, h) - \dot{\hat{\boldsymbol{x}}}_e \\ &= \boldsymbol{A}\hat{\tilde{\boldsymbol{x}}} + \boldsymbol{b}_u\kappa(\hat{\tilde{\boldsymbol{x}}}) + \bar{\boldsymbol{A}}_x\bar{\boldsymbol{e}} - \boldsymbol{\eta}_x(\bar{\boldsymbol{e}}, h) - \dot{\hat{\boldsymbol{x}}}_e\end{aligned} \tag{15}$$

where $\bar{\boldsymbol{A}}_x$ and $\boldsymbol{\eta}_x$ are the first n rows of $\bar{\boldsymbol{A}}$ and $\boldsymbol{\eta}$, respectively. To simplify the expression, (15) can be rewritten as

$$\dot{\hat{\tilde{\boldsymbol{x}}}} = \boldsymbol{A}\hat{\tilde{\boldsymbol{x}}} + \boldsymbol{b}_u\kappa(\hat{\tilde{\boldsymbol{x}}}) + \boldsymbol{\varepsilon} \tag{16}$$

$$\boldsymbol{\varepsilon} = \bar{\boldsymbol{A}}_x\bar{\boldsymbol{e}} - \boldsymbol{\eta}_x(\bar{\boldsymbol{e}}, h) - \dot{\hat{\boldsymbol{x}}}_e. \tag{17}$$

Assumption 5: $\boldsymbol{\sigma}(\cdot) = (\boldsymbol{\sigma}_x(\cdot), \sigma_u(\cdot))$ is continuously differentiable with bounded derivative.

Theorem 1: If the system (1) satisfies Assumptions 1–2, the observer (4) satisfies Assumption 3, the controller (7) satisfies Assumption 4 and the EC (11)–(13) satisfies Assumption 5, then the state of system (1) converges to the equilibrium $\boldsymbol{x}_e = \boldsymbol{\sigma}_x(d)$.

Proof:

i) Assumption 1 ensures that h is bounded and $\lim_{t\to\infty} h = 0$.

ii) $\bar{\boldsymbol{e}}$ is bounded and satisfies $\lim_{t\to\infty} \bar{\boldsymbol{e}} = \boldsymbol{0}$, which can be obtained from Remark 2. This implies $\lim_{t\to\infty} \boldsymbol{e}_x = \boldsymbol{0}$ and $\lim_{t\to\infty} \hat{d} = d$.

iii) Based on Assumption 3, $\boldsymbol{\eta}(\bar{\boldsymbol{e}}, h)$ is bounded and $\lim_{t\to\infty} \boldsymbol{\eta}(\bar{\boldsymbol{e}}, h) = \boldsymbol{0}$.

iv) $\dot{\hat{d}} = \dot{d} - \dot{e}_d = h - \eta_d(\bar{\boldsymbol{e}}, h)$ is bounded and $\lim_{t\to\infty} \dot{\hat{d}} = 0$, where η_d is the last row of $\boldsymbol{\eta}$.

v) Based on Assumption 5, $\dot{\hat{\boldsymbol{x}}}_e = (\mathrm{d}\boldsymbol{\sigma}_x/\mathrm{d}\hat{d})\dot{\hat{d}}$ is bounded and $\lim_{t\to\infty} \dot{\hat{\boldsymbol{x}}}_e = \boldsymbol{0}$.

Therefore, $\boldsymbol{\varepsilon} = \bar{\boldsymbol{A}}_x\bar{\boldsymbol{e}} - \boldsymbol{\eta}_x(\bar{\boldsymbol{e}}, h) - \dot{\hat{\boldsymbol{x}}}_e$ is bounded and $\lim_{t\to\infty} \boldsymbol{\varepsilon} = \boldsymbol{0}$. Based on Assumption 4, system (16) is ISS. Then, by using the property of ISS, we obtain that $\hat{\tilde{\boldsymbol{x}}}$ is bounded and $\lim_{t\to\infty} \hat{\tilde{\boldsymbol{x}}} = \boldsymbol{0}$, implying

$$\begin{aligned}\lim_{t\to\infty} \boldsymbol{x} &= \lim_{t\to\infty} \hat{\tilde{\boldsymbol{x}}} + \hat{\boldsymbol{x}}_e + \boldsymbol{e}_x = \lim_{t\to\infty} \hat{\boldsymbol{x}}_e = \lim_{t\to\infty} \boldsymbol{\sigma}_x(\hat{d}) \\ &= \boldsymbol{\sigma}_x(d) = \boldsymbol{x}_e.\end{aligned}$$

■

E. EC for Accurate Tracking

For accurate tracking, another condition $\lim_{t\to\infty} y_o = 0$ needs to be satisfied. Thus, the equilibrium calculator $\boldsymbol{\sigma}$ should be designed elaborately to meet the following equation:

$$\begin{bmatrix} \boldsymbol{A} & \boldsymbol{b}_u \\ \boldsymbol{c}_o & 0 \end{bmatrix} \begin{bmatrix} \boldsymbol{x}_e \\ u_e \end{bmatrix} = \begin{bmatrix} -\boldsymbol{b}_d \\ 0 \end{bmatrix} d. \tag{18}$$

Theorem 2: If the system (1) satisfies Assumptions 1–2, the observer (4) satisfies Assumption 3, the nominal controller (7) satisfies Assumption 4, then there exists an EC (11)–(13) such that $\lim_{t\to\infty} y_o = 0$ if and only if

$$rank\Bigg(\underbrace{\begin{bmatrix} \boldsymbol{A} & \boldsymbol{b}_u \\ \boldsymbol{c}_o & 0 \end{bmatrix}}_{\boldsymbol{Q}_{uo}}\Bigg) = rank\Bigg(\underbrace{\begin{bmatrix} \boldsymbol{A} & \boldsymbol{b}_u & \boldsymbol{b}_d \\ \boldsymbol{c}_o & 0 & 0 \end{bmatrix}}_{\boldsymbol{Q}_{udo}}\Bigg). \tag{19}$$

Proof: Based on Theorem 1, the following assertions are equivalent: 1) there exists an EC such that $\lim_{t\to\infty} y_o = 0$; 2) (18) has solutions; 3) equation (19) is met. ■

Remark 5: Similar to feedback linearization, which transforms an original nonlinear system to a linear system such that abundant existing methods and conclusions are reusable, ECBC scheme can compensate a broad class of systems with mismatched disturbances (1) to a nominal form (16), and then controllers for nominal systems are still effective and their performances are retained.

Corollary 1: If $\boldsymbol{Q}_{uo}$ is invertible, then the system is compensable and there exists a unique equilibrium

$$\boldsymbol{\sigma}(d) = \begin{bmatrix} \boldsymbol{x}_e^* \\ u_e^* \end{bmatrix} = \boldsymbol{Q}_{uo}^{-1} \begin{bmatrix} -\boldsymbol{b}_d \\ 0 \end{bmatrix} d. \tag{20}$$

Corollary 2: If $\boldsymbol{Q}_{uo}$ is noninvertible and $rank(\boldsymbol{Q}_{uo}) = rank(\boldsymbol{Q}_{udo})$, then the steady output is identically equal to zero for any d, and thus there is no need to compensate.

Proof: As $(\boldsymbol{A}, \boldsymbol{b}_u)$ is controllable, $[\boldsymbol{A} \;\; \boldsymbol{b}_u]$ and $[\boldsymbol{A} \;\; \boldsymbol{b}_u \;\; \boldsymbol{b}_d]$ have full row rank. The conditions of this corollary mean that there exists a unique vector $\boldsymbol{\lambda} \in \mathbb{R}^{1\times n}$ such that $\boldsymbol{\lambda}[\boldsymbol{A} \;\; \boldsymbol{b}_u \;\; \boldsymbol{b}_d] = [\boldsymbol{c}_o \;\; 0 \;\; 0]$. Thus any equilibrium $(\boldsymbol{x}_e, u_e)$ satisfying (10) yields $y_o = \boldsymbol{c}_o\boldsymbol{x}_e = \boldsymbol{\lambda}\boldsymbol{A}\boldsymbol{x}_e = -\boldsymbol{\lambda}\boldsymbol{b}_u - \boldsymbol{\lambda}\boldsymbol{b}_d d = 0$, which implies there is no need to compensate. ■

Corollary 3: If $\boldsymbol{Q}_{uo}$ is non-invertible and $rank(\boldsymbol{Q}_{uo}) \neq rank(\boldsymbol{Q}_{udo})$, then any equilibrium does not influence the steady output, i.e., neither cancel out nor decrease the effect of disturbance, and thus the system cannot be compensated.

Proof: Let $\boldsymbol{A}_u = [\boldsymbol{A} \;\; \boldsymbol{b}_u]$, $\boldsymbol{c}_{ou} = [c_o \;\; 0]$. The fact that $\boldsymbol{Q}_{uo}$ is noninvertible implies $\boldsymbol{c}_{ou}$ lies in the space spanned by the rows of $\boldsymbol{A}_u$, i.e., there exists a unique vector $\boldsymbol{\lambda} = \boldsymbol{c}_{ou}\boldsymbol{A}_u^T(\boldsymbol{A}_u\boldsymbol{A}_u^T)^{-1} \neq \boldsymbol{0}$ such that $\boldsymbol{\lambda}[\boldsymbol{A} \;\; \boldsymbol{b}_u] = [\boldsymbol{c}_o \;\; 0]$. $rank(\boldsymbol{Q}_{uo}) \neq rank(\boldsymbol{Q}_{udo})$ means $\boldsymbol{\lambda}\boldsymbol{b}_d \neq \boldsymbol{0}$. Therefore, any equilibrium $(\boldsymbol{x}_e, u_e)$ satisfying (10) yields $y_o = \boldsymbol{c}_o\boldsymbol{x}_e = \boldsymbol{\lambda}\boldsymbol{A}\boldsymbol{x}_e + \boldsymbol{\lambda}\boldsymbol{b}_u = -\boldsymbol{\lambda}\boldsymbol{b}_d d \neq 0$, which implies the system cannot be compensated. ■

Remark 6: Theorems 1–2 showcase the steady-state characteristic of ECBC, illustrating that the impact of a disturbance can be fully counteracted under specific conditions. Moreover, the ECBC scheme also demonstrates excellent dynamic performance if the nominal controller is appropriately selected or optimized. Based on (16) and (17), if the initial estimation error satisfies $\bar{\boldsymbol{e}}_0 = 0$ and the disturbance d is constant, then $\bar{\boldsymbol{e}} \equiv \boldsymbol{0}$, such that the closed-loop system of ECBC (16) is reduced to the nominal closed-loop system (9) with $\boldsymbol{\varepsilon} = 0$ and hence the performance of the nominal controller is retained.

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III. ECBC for MIMO Systems

A. System Formulation

Most results of ECBC for SISO systems still hold for MIMO systems with several extensions. Moreover, the complexity of MIMO systems brings about several new meaningful and interesting properties.

Consider an MIMO system with multiple disturbances

$$\begin{aligned}\dot{\boldsymbol{x}} &= \boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}_u\boldsymbol{u} + \boldsymbol{B}_d\boldsymbol{d}\\ \boldsymbol{y}_m &= \boldsymbol{C}_m\boldsymbol{x}\\ \boldsymbol{y}_o &= \boldsymbol{C}_o\boldsymbol{x}\end{aligned} \tag{21}$$

where $\boldsymbol{x} \in \mathbb{R}^n, \boldsymbol{u} \in \mathbb{R}^{n_u}, \boldsymbol{d} \in \mathbb{R}^{n_d}, \boldsymbol{y}_m \in \mathbb{R}^{n_m}, \boldsymbol{y}_o \in \mathbb{R}^{n_o}$, and the matrices $\boldsymbol{A}, \boldsymbol{B}_u, \boldsymbol{B}_d, \boldsymbol{C}_m, \boldsymbol{C}_o$ have appropriate dimensions. Without loss of generality we assume that $\boldsymbol{B}_u, \boldsymbol{B}_d$ have full column rank and $\boldsymbol{C}_m, \boldsymbol{C}_o$ have full row rank. Besides, we assume that Assumptions 1–4 still hold.

In MIMO systems, Lemma 1 still holds with modified $\boldsymbol{Q}_{dm}$

$$\boldsymbol{Q}_{dm} = \begin{bmatrix}\boldsymbol{A} & \boldsymbol{B}_d\\ \boldsymbol{C}_m & \boldsymbol{0}\end{bmatrix}. \tag{22}$$

Corollary 4: Consider an observable system $(\boldsymbol{A}, \boldsymbol{C}_m)$. If there are independent disturbances in all state channels, i.e., $rank(\boldsymbol{B}_d) = n$, then the extended system $(\bar{\boldsymbol{A}}, \bar{\boldsymbol{C}}_m)$ is observable if and only if $rank(\boldsymbol{C}_m) = n$.

Corollary 5: Consider an observable system $(\boldsymbol{A}, \boldsymbol{C}_m)$ with general disturbance satisfying $rank(\boldsymbol{B}_d) = n_d$. A necessary condition for that $(\bar{\boldsymbol{A}}, \bar{\boldsymbol{C}}_m)$ is observable is $rank(\boldsymbol{C}_m) \geq n_d$.

The proofs of Corollaries 4–5 are similar to Lemma 1, so they are omitted due to the space limitations.

B. Decomposition of Controlled Output

Theorem 3: If Assumptions 1–4 are met, there exists an EC (11)–(13) such that $\lim_{t\to\infty} \boldsymbol{y}_o = 0$ if and only if

$$rank\Bigg(\underbrace{\begin{bmatrix}\boldsymbol{A} & \boldsymbol{B}_u\\ \boldsymbol{C}_o & 0\end{bmatrix}}_{\boldsymbol{Q}_{uo}}\Bigg) = rank\Bigg(\underbrace{\begin{bmatrix}\boldsymbol{A} & \boldsymbol{B}_u & \boldsymbol{B}_d\\ \boldsymbol{C}_o & 0 & 0\end{bmatrix}}_{\boldsymbol{Q}_{udo}}\Bigg). \tag{23}$$

The proof is similar to Theorem 2 and thus is omitted.

Different from SISO systems, there exist cases in MIMO systems that the effects of disturbances on some outputs can be canceled, while the others cannot. Given an MIMO system, we concern which channels can be compensated, how many solutions there exist, and what is the optimal solution when the effects cannot be completely compensated.

To answer these questions, a possible solution is to decompose the controlled output matrix $\boldsymbol{C}_o$ by the properties of its each row. $\boldsymbol{C}_o$ can be divided into three portions as

$$\boldsymbol{C}_o = \begin{bmatrix}\boldsymbol{C}_{o,i}^T & \boldsymbol{C}_{o,j}^T & \boldsymbol{C}_{o,k}^T\end{bmatrix}^T \tag{24}$$

where the numbers of rows of $\boldsymbol{C}_{o,i}$, $\boldsymbol{C}_{o,j}$, $\boldsymbol{C}_{o,k}$ are $n_{o,i}$, $n_{o,j}$, $n_{o,k}$, respectively. The definitions and corresponding properties of each portion are as follows.

First, we introduce the following notations for brevity:

$$\boldsymbol{Q}_{uo,r} = \begin{bmatrix}\boldsymbol{A} & \boldsymbol{B}_u\\ \boldsymbol{C}_{o,r} & 0\end{bmatrix}, \boldsymbol{Q}'_{uo,r} = \begin{bmatrix}\boldsymbol{A} & \boldsymbol{B}_u\\ \boldsymbol{c}_{o,r} & 0\end{bmatrix}$$

$$\boldsymbol{Q}_{udo,r} = \begin{bmatrix}\boldsymbol{A} & \boldsymbol{B}_u & \boldsymbol{B}_d\\ \boldsymbol{C}_{o,r} & 0 & 0\end{bmatrix}, r \in \{i, j, k\}.$$

1) Compensable portion $\boldsymbol{C}_{o,i}$: Each row $\boldsymbol{c}_{o,i}$ satisfies that $\boldsymbol{Q}'_{uo,i}$ has full row rank, which implies that

$$n < rank(\boldsymbol{Q}_{uo,i}) \leq rank(\boldsymbol{Q}_{udo,i}) \leq n + n_{o,i}. \tag{25}$$

Different from SISO system, even though each row $\boldsymbol{c}_{o,i}$ satisfies that $\boldsymbol{Q}'_{uo,i}$ has full row rank, when it comes to the whole matrix $\boldsymbol{C}_{o,i}$, the matrix $\boldsymbol{Q}_{uo,i}$ may not have full row rank. Namely, that each output channel is compensable separately, does not mean all channels can be compensated simultaneously. An example of this case is

$$\boldsymbol{A} = \begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}, \boldsymbol{B}_u = \begin{bmatrix}1\\ 1\\ 1\end{bmatrix}, \boldsymbol{B}_d = \begin{bmatrix}0\\ 1\\ 1\end{bmatrix}, \boldsymbol{C}_o = \begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}.$$

Starting from this new property, the next Section III-C presents compensation methods for this portion.

2) No-Need-to-Compensate Portion $\boldsymbol{C}_{o,j}$: Each row $\boldsymbol{c}_{o,j}$ satisfies that $\boldsymbol{Q}'_{uo,j}$ is not of full row rank and $rank(\boldsymbol{Q}'_{uo,j}) = rank(\boldsymbol{Q}'_{udo,j})$, which implies

$$rank(\boldsymbol{Q}_{uo,j}) = rank(\boldsymbol{Q}_{udo,j}) = n. \tag{26}$$

There exists a unique matrix $\boldsymbol{\lambda}_{o,j} = \boldsymbol{C}_{ou,j}\boldsymbol{A}_u^T(\boldsymbol{A}_u\boldsymbol{A}_u^T)^{-1} \neq \boldsymbol{0}$ such that $\boldsymbol{\lambda}[\boldsymbol{A} \quad \boldsymbol{B}_u] = [\boldsymbol{C}_{o,j} \quad \boldsymbol{0}]$, and $\boldsymbol{\lambda}\boldsymbol{B}_d = \boldsymbol{0}$. Hence any equilibrium $(\boldsymbol{x}_e, \boldsymbol{u}_e)$ satisfying (10) yields $\boldsymbol{y}_o = \boldsymbol{C}_o\boldsymbol{x}_e = \boldsymbol{\lambda}\boldsymbol{A}\boldsymbol{x}_e = -\boldsymbol{\lambda}\boldsymbol{B}_u - \boldsymbol{\lambda}\boldsymbol{B}_d\boldsymbol{d} = 0$, implying that there is no need to compensate this portion.

3) Cannot-be-Compensated Portion $\boldsymbol{C}_{o,k}$: Each row $\boldsymbol{c}_{o,k}$ satisfies that $\boldsymbol{Q}'_{uo,k}$ is not of full row rank and $rank(\boldsymbol{Q}'_{uo,j}) < rank(\boldsymbol{Q}'_{udo,j})$, which implies

$$n = rank(\boldsymbol{Q}_{uo,k}) < rank(\boldsymbol{Q}_{udo,k}) \leq n + n_{o,k}. \tag{27}$$

There exists a unique matrix $\boldsymbol{\lambda}_{o,k} = \boldsymbol{C}_{ou,k}\boldsymbol{A}_u^T(\boldsymbol{A}_u\boldsymbol{A}_u^T)^{-1} \neq \boldsymbol{0}$ such that $\boldsymbol{\lambda}[\boldsymbol{A} \quad \boldsymbol{B}_u] = [\boldsymbol{C}_{o,k} \quad \boldsymbol{0}]$ and $\boldsymbol{\lambda}\boldsymbol{B}_d \neq \boldsymbol{0}$. Hence any equilibrium $(\boldsymbol{x}_e, \boldsymbol{u}_e)$ satisfying (10) yields $\boldsymbol{y}_o = \boldsymbol{C}_o\boldsymbol{x}_e = \boldsymbol{\lambda}\boldsymbol{A}\boldsymbol{x}_e = -\boldsymbol{\lambda}\boldsymbol{B}_u - \boldsymbol{\lambda}\boldsymbol{B}_d\boldsymbol{d} \neq 0$, implying that this portion cannot be compensated.

C. Compensability

Focusing on the compensable portion $\boldsymbol{C}_{o,i}$, this subsection analyzes the number of equilibrium solutions. When the system cannot be completely compensated, a minimum norm solution is supplied. For simplicity, the subscript i is omitted in the follows.

Corollary 6 (Complete Compensation With a Unique Solution): If $rank(\boldsymbol{Q}_{uo}) = rank(\boldsymbol{Q}_{udo}) = n + n_u$, there exists a unique solution

$$\boldsymbol{\sigma}(\boldsymbol{d}) = \begin{bmatrix}\boldsymbol{x}_e^*\\ \boldsymbol{u}_e^*\end{bmatrix} = \boldsymbol{Q}_{uo}^{-1}\begin{bmatrix}\boldsymbol{B}_d\\ \boldsymbol{0}\end{bmatrix}\boldsymbol{d}. \tag{28}$$

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Corollary 7 (Complete Compensation With Multiple Solutions): If $rank(\boldsymbol{Q}_{uo}) = rank(\boldsymbol{Q}_{udo}) < n + n_u$, there exists a class of infinitely many solutions in the following form:

$$\boldsymbol{\sigma}(\boldsymbol{d}) = \begin{bmatrix} \boldsymbol{x}_e^* \\ \boldsymbol{u}_e^* \end{bmatrix} = \boldsymbol{Q}_{uo}^+ \begin{bmatrix} \boldsymbol{B}_d \\ \boldsymbol{0} \end{bmatrix} \boldsymbol{d} + (\boldsymbol{I} - \boldsymbol{Q}_{uo}^+ \boldsymbol{Q}_{uo})\boldsymbol{w} \tag{29}$$

where $\boldsymbol{Q}_{uo}^+ = \boldsymbol{Q}_{uo}^T(\boldsymbol{Q}_{uo}\boldsymbol{Q}_{uo}^T)^{-1}$, and $\boldsymbol{w}$ is a free vector.

Corollary 8 (Minimum Weighted Norm Compensation): If $rank(\boldsymbol{Q}_{uo}) < rank(\boldsymbol{Q}_{udo})$, then all the output channels cannot be completely compensated simultaneously. Thus, the minimum P-weighted norm solution can be obtained by solving the following optimization problem:

$$\begin{aligned} \min_{\boldsymbol{x}_e, \boldsymbol{u}_e} \quad & \|\boldsymbol{C}_o \boldsymbol{x}_e\|_{\boldsymbol{P}} \\ \text{s.t.} \quad & \boldsymbol{A}\boldsymbol{x}_e + \boldsymbol{B}_u \boldsymbol{u}_e = -\boldsymbol{B}_d \boldsymbol{d} \end{aligned} \tag{30}$$

where $\boldsymbol{P}$ is a symmetric positive definite matrix and the P-weighted norm is defined as $\|\boldsymbol{C}_o \boldsymbol{x}_e\|_{\boldsymbol{P}} = (\boldsymbol{x}_e^T \boldsymbol{C}_o^T \boldsymbol{P} \boldsymbol{C}_o \boldsymbol{x}_e)^{1/2}$.

Problem (30) is a convex quadratic programming (QP) problem, and the solutions satisfy the following KKT condition, in which $\boldsymbol{\nu}$ is a Lagrangian multiplier:

$$\begin{bmatrix} \boldsymbol{C}_o^T \boldsymbol{P} \boldsymbol{C}_o & \boldsymbol{0} & \boldsymbol{A}^T \\ \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{B}_u^T \\ \boldsymbol{A} & \boldsymbol{B}_u & \boldsymbol{0} \end{bmatrix} \begin{bmatrix} \boldsymbol{x}_e \\ \boldsymbol{u}_e \\ \boldsymbol{\nu} \end{bmatrix} = - \begin{bmatrix} \boldsymbol{0} \\ \boldsymbol{0} \\ \boldsymbol{B}_d \end{bmatrix} \boldsymbol{d}. \tag{31}$$

Remark 7: If $rank(\boldsymbol{Q}_{uo}) = n + n_u$, then KKT system (31) has a unique solution. If $rank(\boldsymbol{Q}_{uo}) < n + n_u$, then (31) has infinitely many solutions. In fact, both (28) and (29) are solutions to (30), too.

IV. ECBC for Systems With Sensor Errors

A. System Formulation

In physical systems, apart from the disturbances in the state channels, there may exist sensor errors in the measurable output channels, whose mean values are nonzero. For instance, the roll channel of the IMU (inertial measurement unit) in the robot of this paper has an offset that cannot be completely removed by calibrating in advance because of the effects of terrains and payloads. Therefore, this section deals with systems with state disturbances and sensor errors described as

$$\begin{aligned} \dot{\boldsymbol{x}} &= \boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}_u \boldsymbol{u} + \boldsymbol{B}_d \boldsymbol{d} \\ \boldsymbol{y}_m &= \boldsymbol{C}_m \boldsymbol{x} + \boldsymbol{E}_m \boldsymbol{g} \\ \boldsymbol{y}_o &= \boldsymbol{C}_o \boldsymbol{x} \end{aligned} \tag{32}$$

where $\boldsymbol{g} \in \mathbb{R}^{n_g}$ denotes a constant sensor error. Without loss of generality we assume that $\boldsymbol{B}_u, \boldsymbol{B}_d, \boldsymbol{E}_m$ have full column rank and $\boldsymbol{C}_m, \boldsymbol{C}_o$ have full row rank.

Since $\boldsymbol{C}_m$ is of full row rank, there exists a constant vector $\boldsymbol{x}_g = \boldsymbol{C}_m^+ \boldsymbol{E}_m \boldsymbol{g}$ such that $\boldsymbol{C}_m \boldsymbol{x}_g = \boldsymbol{E}_m \boldsymbol{g}$, where $\boldsymbol{C}_m^+ = \boldsymbol{C}_m^T(\boldsymbol{C}_m \boldsymbol{C}_m^T)^{-1}$ denotes the Moore–Penrose inverse of $\boldsymbol{C}_m$. Let $\boldsymbol{v} = \boldsymbol{x} + \boldsymbol{x}_g$ denotes a new state, by taking the derivative of $\boldsymbol{v}$, we have

$$\begin{aligned} \dot{\boldsymbol{v}} &= \dot{\boldsymbol{x}} + \dot{\boldsymbol{x}}_g = \boldsymbol{A}(\boldsymbol{v} - \boldsymbol{x}_g) + \boldsymbol{B}_u \boldsymbol{u} + \boldsymbol{B}_d \boldsymbol{d} \\ &= \boldsymbol{A}\boldsymbol{v} + \boldsymbol{B}_u \boldsymbol{u} + \boldsymbol{B}_d \boldsymbol{d} - \boldsymbol{A}\boldsymbol{C}_m^+ \boldsymbol{E}_m \boldsymbol{g} \\ \boldsymbol{y}_m &= \boldsymbol{C}_m(\boldsymbol{v} - \boldsymbol{x}_g) + \boldsymbol{E}_m \boldsymbol{g} = \boldsymbol{C}_m \boldsymbol{v} \\ \boldsymbol{y}_o &= \boldsymbol{C}_o(\boldsymbol{v} - \boldsymbol{x}_g) = \boldsymbol{C}_o \boldsymbol{v} - \boldsymbol{C}_o \boldsymbol{C}_m^+ \boldsymbol{E}_m \boldsymbol{g}. \end{aligned} \tag{33}$$

For the sake of simplicity, (33) can be rewritten as

$$\begin{aligned} \dot{\boldsymbol{v}} &= \boldsymbol{A}\boldsymbol{v} + \boldsymbol{B}_u \boldsymbol{u} + \boldsymbol{B}_{dg} \boldsymbol{d}_g \\ \boldsymbol{y}_m &= \boldsymbol{C}_m \boldsymbol{v} \\ \boldsymbol{y}_o &= \boldsymbol{C}_o \boldsymbol{v} + \boldsymbol{E}_o \boldsymbol{g} \end{aligned} \tag{34}$$

where $\boldsymbol{B}_g = -\boldsymbol{A}\boldsymbol{C}_m^+ \boldsymbol{E}_m$, $\boldsymbol{B}_{dg} = [\boldsymbol{B}_d \ \boldsymbol{B}_g]$, $\boldsymbol{d}_g = [\boldsymbol{d}^T \ \boldsymbol{g}^T]^T$ and $\boldsymbol{E}_o = -\boldsymbol{C}_o \boldsymbol{C}_m^+ \boldsymbol{E}_m$.

The above transform suggests that the sensor errors can be converted to additional disturbances in the state channels. Define $\boldsymbol{Q}_{dgm}$, $\bar{\boldsymbol{Q}}_{dgm}$, and $\bar{\boldsymbol{Q}}_{udgo}$ as

$$\boldsymbol{Q}_{dgm} = \begin{bmatrix} \boldsymbol{A} & \boldsymbol{B}_d & \boldsymbol{0} \\ \boldsymbol{C}_m & \boldsymbol{0} & \boldsymbol{E}_m \end{bmatrix}, \quad \bar{\boldsymbol{Q}}_{dgm} = \begin{bmatrix} \boldsymbol{A} & \boldsymbol{B}_{dg} \\ \boldsymbol{C}_m & \boldsymbol{0} \end{bmatrix}$$

$$\boldsymbol{Q}_{udgo} = \begin{bmatrix} \boldsymbol{A} & \boldsymbol{B}_u & \boldsymbol{B}_{dg} \\ \boldsymbol{C}_o & \boldsymbol{0} & \boldsymbol{0} \end{bmatrix}.$$

It is possible to claim the observability results by Lemma 3.

Lemma 3: The extended system corresponding to system (32) is observable, if and only if $(\boldsymbol{A}, \boldsymbol{C}_m)$ is observable and $\boldsymbol{Q}_{dgm}$ has full column rank.

Proof: This lemma can be obtained from Lemma 1 and the following equation:

$$\bar{\boldsymbol{Q}}_{dgm} = \boldsymbol{Q}_{dgm} \begin{bmatrix} \boldsymbol{I} & \boldsymbol{0} & -\boldsymbol{C}_m^+ \boldsymbol{E}_m \\ \boldsymbol{0} & \boldsymbol{I} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{I} \end{bmatrix}.$$

■

Remark 8: That system $(\bar{\boldsymbol{A}}, \bar{\boldsymbol{C}}_m)$ is not observable only means that the disturbances and sensor errors cannot be observed independently, but combinations of them could be observable.

B. Compensability

It can be noticed that if $\boldsymbol{Q}_{dgm}$ has full column rank and $\boldsymbol{E}_o = 0$, then system (34) is reduced to system (21), and the results in Section III still hold. If $\boldsymbol{Q}_{dgm}$ is not of full column rank, we have the following theorem:

Theorem 4: If system (34) meets the following conditions:

1) $rank(\bar{\boldsymbol{Q}}_{dgm}) = n + rank(\boldsymbol{B}_{dg})$.
2) $rank(\boldsymbol{Q}_{uo}) = rank(\boldsymbol{Q}_{udgo})$.
3) $\boldsymbol{E}_o = \boldsymbol{0}$.

Then there exists an EC such that the effects of disturbances and sensor errors can be removed from the controlled outputs.

Proof: Assume $rank(\boldsymbol{B}_{dg}) = n_r$. Matrix $\boldsymbol{B}_{dg}$ can be divided into to two blocks: the one $\boldsymbol{B}_{df}$ is constructed by selecting n_r linearly independent columns, and the other $\boldsymbol{B}_{dr}$ is constructed by the remaining columns. The rows of $\boldsymbol{d}$ corresponding to $\boldsymbol{B}_{df}$ and $\boldsymbol{B}_{dr}$ are $\boldsymbol{d}_f$ and $\boldsymbol{d}_r$. There exists

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$\boldsymbol{\lambda} \in \mathbb{R}^{n_r \times (n_d+n_g-n_r)}$ such that $\boldsymbol{B}_{dr} = \boldsymbol{B}_{df}\boldsymbol{\lambda}$. Then $\boldsymbol{B}_{dg}\boldsymbol{d} = \boldsymbol{B}_{df}(\boldsymbol{d}_f + \boldsymbol{\lambda}\boldsymbol{d}_r)$. Let $\boldsymbol{B}'_d = \boldsymbol{B}_{df}$ and $\boldsymbol{d}' = \boldsymbol{d}_f + \boldsymbol{\lambda}\boldsymbol{d}_r$. Considering condition 3), system (34) can be rewritten as

$$\begin{aligned} \dot{\boldsymbol{v}} &= \boldsymbol{A}\boldsymbol{v} + \boldsymbol{B}_u\boldsymbol{u} + \boldsymbol{B}'_d\boldsymbol{d}' \\ \boldsymbol{y}_m &= \boldsymbol{C}_m\boldsymbol{v} \\ \boldsymbol{y}_o &= \boldsymbol{C}_o\boldsymbol{v} \end{aligned} \tag{35}$$

where we have

$$rank\left(\begin{bmatrix} \boldsymbol{A} & \boldsymbol{B}'_d \\ \boldsymbol{C}_m & \boldsymbol{0} \end{bmatrix}\right) = rank(\bar{\boldsymbol{Q}}_{dgm}) = n + n_r$$

$$rank(\boldsymbol{Q}_{uo}) = rank(\boldsymbol{Q}_{udgo}) = rank\left(\begin{bmatrix} \boldsymbol{A} & \boldsymbol{B}_u & \boldsymbol{B}'_d \\ \boldsymbol{C}_o & \boldsymbol{0} & \boldsymbol{0} \end{bmatrix}\right).$$

The result of Theorem 4 can be obtained from Theorem 3. ■

V. Relations With Existing Methods

A. GESOBC

The ECBC scheme is significantly inspired by GESOBC [10], and the two methods exhibit the following relations.

1) ECBC improves the solvability condition provided by GESOBC. Based on [10, Th. 2], the solvability condition of GESOBC for SISO systems is that "the feedback control gain $\boldsymbol{K}_x$ is chosen such that $\boldsymbol{A}_f = \boldsymbol{A} + \boldsymbol{b}_u\boldsymbol{K}_x$ is Hurwitz and $\boldsymbol{c}_o\boldsymbol{A}_f^{-1}\boldsymbol{b}_u$ is invertible". It seems that the solvability depends on the control gain. Namely, to determine whether $\boldsymbol{c}_o\boldsymbol{A}_f^{-1}\boldsymbol{b}_u$ is invertible, it is necessary to design the control gain $\boldsymbol{K}_x$ in advance.

Fortunately, we discover that the solvability condition can be reformulated to a new form that is independent of the control gain, as shown by the following statement.

Corollary 9: That $\boldsymbol{c}_o\boldsymbol{A}_f^{-1}\boldsymbol{b}_u$ is invertible is equivalent to that $\boldsymbol{Q}_{uo}$ is invertible (the condition in Corollary 1).

Proof: First, we have

$$\underbrace{\begin{bmatrix} \boldsymbol{A}_f & \boldsymbol{b}_u \\ \boldsymbol{c}_o & 0 \end{bmatrix}}_{\boldsymbol{Q}_f} = \begin{bmatrix} \boldsymbol{A} + \boldsymbol{b}_u\boldsymbol{K}_x & \boldsymbol{b}_u \\ \boldsymbol{c}_o & 0 \end{bmatrix} = \underbrace{\begin{bmatrix} \boldsymbol{A} & \boldsymbol{b}_u \\ \boldsymbol{c}_o & 0 \end{bmatrix}}_{\boldsymbol{Q}_{uo}} \underbrace{\begin{bmatrix} \boldsymbol{I}_n & \boldsymbol{0} \\ \boldsymbol{K}_x & 1 \end{bmatrix}}_{\boldsymbol{L}} \tag{36}$$

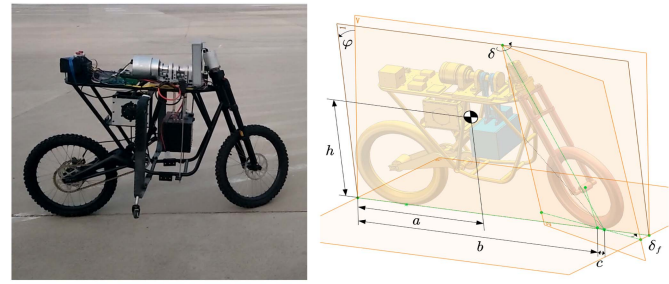
where $\boldsymbol{I}_n$ is an identity matrix of size n. Notice that the Schur complement of $\boldsymbol{A}_f$ in $\boldsymbol{Q}_f$ is $-\boldsymbol{c}_o\boldsymbol{A}_f^{-1}\boldsymbol{b}_u$. Since $\boldsymbol{A}_f$ is invertible, and according to the property of Schur complement, that $\boldsymbol{c}_o\boldsymbol{A}_f^{-1}\boldsymbol{b}_u$ is invertible is equivalent to that $\boldsymbol{Q}_f$ is invertible. Given that $\boldsymbol{L}$ is invertible independently of $\boldsymbol{K}_x$, that $\boldsymbol{Q}_f$ is invertible is equivalent to that $\boldsymbol{Q}_{uo}$ is invertible. ■

Remark 9: Corollary 9 suggests that whether the mismatched disturbances can be completely compensated only depends on the open-loop system defined by $\boldsymbol{A}, \boldsymbol{b}_u, \boldsymbol{c}_o$.

2) GESOBC is a special case of ECBC that employs a state feedback controller, i.e., $\kappa(\hat{\tilde{\boldsymbol{x}}}) = \boldsymbol{K}_x\hat{\tilde{\boldsymbol{x}}}$. Based on (20) and (36), the controller (13) can be reformulated as

$$\begin{aligned} u = \kappa(\hat{\tilde{\boldsymbol{x}}}) + \hat{u}_e &= \boldsymbol{K}_x\hat{\boldsymbol{x}} + \begin{bmatrix} -\boldsymbol{K}_x & 1 \end{bmatrix} \begin{bmatrix} \hat{\boldsymbol{x}}_e^T & \hat{u}_e^T \end{bmatrix}^T \\ &= \boldsymbol{K}_x\hat{\boldsymbol{x}} + \begin{bmatrix} -\boldsymbol{K}_x & 1 \end{bmatrix} \boldsymbol{Q}_{uo}^{-1} \begin{bmatrix} \boldsymbol{b}_d^T & 0 \end{bmatrix}^T \hat{d} \\ &= \boldsymbol{K}_x\hat{\boldsymbol{x}} + \begin{bmatrix} -\boldsymbol{K}_x & 1 \end{bmatrix} \boldsymbol{L}\boldsymbol{Q}_f^{-1} \begin{bmatrix} \boldsymbol{b}_d^T & 0 \end{bmatrix}^T \hat{d} \\ &= \boldsymbol{K}_x\hat{\boldsymbol{x}} + (\boldsymbol{c}_o\boldsymbol{A}_f^{-1}\boldsymbol{b}_d)^{-1}\boldsymbol{c}_o\boldsymbol{A}_f^{-1}\boldsymbol{b}_d\hat{d} \end{aligned}$$

which is equivalent to the GESOBC.



(a) (b)

Fig. 2. (a) STTW robot prototype. (b) Schematic of the STTW robot.

B. Offset-Free MPC

The relations between ECBC and offset-free MPC [17] are listed as follows.

1) In offset-free MPC, the number of disturbances and the disturbance model $\boldsymbol{B}_d, \boldsymbol{C}_g$ are designable, while $\boldsymbol{B}_d$ and $\boldsymbol{C}_g$ in ECBC is defined by modeling.
2) In [17], the controlled outputs are linear combinations of measurable outputs, i.e. $\boldsymbol{y}_o = \boldsymbol{H}\boldsymbol{y}_m$. Since $\boldsymbol{y}_m = \boldsymbol{C}_m\boldsymbol{x} + \boldsymbol{C}_g\boldsymbol{g}$, there are sensor errors in the controlled outputs $\boldsymbol{y}_o$, so the "offset-free" is in terms of the perturbed outputs.

C. DOB-SMC

The relations between ECBC and DOB-SMC include the following.

1) If one selects SMC/NTSMC (nonsingular terminal SMC) as controller (7) and NDOB/FTDO (finite-time DOB) as observer (4), then the resulting ECBC is equivalent to the methods in [8] and [9].
2) The plants in DOB-SMC are limited to chain of integrators, resulting that the method cannot be used in nonminimum phase systems, while ECBC can be applied to such systems based on existing methods for disturbance-free systems without any special modifications. The experiments in the Section VI demonstrate this feature.

VI. Example and Experiments

In this section, an STTW robot, as shown in Fig. 2(a), is selected as a practical example to demonstrate the design procedures of ECBC, where four nominal controllers are migrated to

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the perturbed system. The simulation and experimental results validate the ECBC's capability of disturbance rejection.

A. STTW Robot Model

The nonlinear dynamics of an STTW robot from [30] with slight modifications is described as follows:

$$\begin{aligned}&(mh^2+I_b)\ddot{\varphi}-mgh\sin\varphi+\frac{mah\cos\lambda}{b}(\dot{\delta}v_x+\delta\dot{v}_x))\\&-\frac{macg\cos\lambda}{b\cos^2\varphi}\delta+\frac{mhv_x^2\cos\lambda}{b}\delta\left(1-\frac{h\delta\cos\lambda\tan\varphi}{b}\right)\\&+\frac{mahv_x\cos\lambda\tan\varphi}{b}\delta\dot{\varphi}=0.\end{aligned}\tag{37}$$

where $v_x, \varphi, \delta, \lambda, m, I_b$ denote forward velocity, roll angle, steering angle, caster angle, mass, moment of inertial about the longitudinal axis, respectively, and other geometric parameters are depicted in Fig. 2(b). The state is chosen as $\boldsymbol{x}=[\delta \quad \varphi \quad \dot{\varphi}]^T$ and the input is $u=\dot{\delta}$. Linearized around the origin, the matrices in system (21) are derived as follows:

$$\begin{aligned}&\boldsymbol{A}=\begin{bmatrix}0&0&0\\0&0&1\\a_1&a_2&0\end{bmatrix},\boldsymbol{B}_u=\begin{bmatrix}1\\0\\b_1\end{bmatrix},\boldsymbol{B}_d=\begin{bmatrix}0\\0\\1\end{bmatrix}\\&\boldsymbol{C}_m=\begin{bmatrix}1&0&0\\0&1&0\end{bmatrix},\boldsymbol{E}_m=\begin{bmatrix}0\\1\end{bmatrix}\end{aligned}\tag{38}$$

where a_1, a_2, and b_1 are nonzero coefficients and $a_1b_1>0$. The matrix $\boldsymbol{B}_d$ shows that there exists a unique disturbance in the $\dot{\varphi}$ channel, including the residual nonlinear dynamics, parameter perturbation and external torque. According to Definition 1, that $\boldsymbol{B}_u$ and $\boldsymbol{B}_d$ are linearly independent implies that the disturbance is mismatched. The matrix $\boldsymbol{E}_m$ indicates that only the roll channel has sensor error. $\boldsymbol{B}_g$ can be obtained by

$$\boldsymbol{B}_g=-\boldsymbol{A}\boldsymbol{C}_m^T(\boldsymbol{C}_m\boldsymbol{C}_m^T)^{-1}\boldsymbol{E}_m=\begin{bmatrix}0&0&-a_2\end{bmatrix}^T\tag{39}$$

and the three controlled output matrices are defined as follows:

$$\boldsymbol{C}_{o,1}=[1\quad 0\quad 0],\ \boldsymbol{C}_{o,2}=[0\quad 1\quad 0],\ \boldsymbol{C}_{o,3}=[0\quad 0\quad 1].$$

B. Compensability Analysis

1) Observability of the Disturbance and Sensor Error: Consider the ranks of the following matrices:

$$\begin{aligned}&\boldsymbol{Q}_1=\begin{bmatrix}\boldsymbol{A}&\boldsymbol{0}\\\boldsymbol{C}_m&\boldsymbol{E}_m\end{bmatrix},\boldsymbol{Q}_2=\begin{bmatrix}\boldsymbol{A}&\boldsymbol{B}_d\\\boldsymbol{C}_m&\boldsymbol{0}\end{bmatrix}\\&\boldsymbol{Q}_3=\begin{bmatrix}\boldsymbol{A}&\boldsymbol{B}_d&\boldsymbol{0}\\\boldsymbol{C}_m&\boldsymbol{0}&\boldsymbol{E}_m\end{bmatrix}.\end{aligned}\tag{40}$$

We have

- $rank(\boldsymbol{Q}_1)=4$. The extended system only with sensor error is observable.
- $rank(\boldsymbol{Q}_2)=4$. The extended system only with disturbance is observable.
- $rank(\boldsymbol{Q}_3)=4<5$. The disturbance and sensor error cannot be observed independently.
- $rank(\boldsymbol{B}_{dg})=1$ and $rank(\boldsymbol{Q}_3)=3+rank(\boldsymbol{B}_{dg})$. According to Theorem 4, there exists $\boldsymbol{B}_d'=\boldsymbol{B}_d, \boldsymbol{d}'=d-a_3g$ such that the combined disturbance is observable.

2) Compensability of the Controlled Output Channels: Consider the following matrices:

$$\begin{aligned}&\boldsymbol{Q}_{uo,i}=\begin{bmatrix}\boldsymbol{A}&\boldsymbol{B}_u\\\boldsymbol{C}_{o,i}&\boldsymbol{0}\end{bmatrix},\boldsymbol{Q}_{udo,i}=\begin{bmatrix}\boldsymbol{A}&\boldsymbol{B}_u&\boldsymbol{B}_d'\\\boldsymbol{C}_{o,i}&\boldsymbol{0}&\boldsymbol{0}\end{bmatrix}\\&\boldsymbol{E}_{o,i}=-\boldsymbol{C}_{o,i}\boldsymbol{C}_m^T(\boldsymbol{C}_m\boldsymbol{C}_m^T)^{-1}\boldsymbol{E}_m,\quad i\in\{1,2,3\}.\end{aligned}\tag{41}$$

The following properties are derived:

- For the first controlled output (steering angle), $\boldsymbol{E}_{o,1}=0$, $rank(\boldsymbol{Q}_{uo,1})=rank(\boldsymbol{Q}_{udo,1})=4$, which means this channel is compensable.
- For the second controlled output (roll angle), $rank(\boldsymbol{Q}_{uo,2})=rank(\boldsymbol{Q}_{udo,2})=4$ but $\boldsymbol{E}_{o,2}\neq 0$, from which we cannot determine whether this channel can be compensated since Theorem 4 just provides sufficient conditions. Indeed, if there is only sensor error, then this channel is compensable.
- For the third controlled output (roll angular rate), $\boldsymbol{E}_{o,3}=0, rank(\boldsymbol{Q}_{uo,3})=rank(\boldsymbol{Q}_{udo,1})=3$, which means it is unnecessary to compensate this channel.

From a practical view, the controller's role is to follow the trajectories from the planner, so we select the steering angle as the controlled output. For simplicity, let $\boldsymbol{C}_o=\boldsymbol{C}_{o,1}$.

C. Controllers Design

In the simulation and experiments, we choose GESO [10] as the DOB and choose four controllers, including an H_∞ controller, an MPC controller, a state feedback controller by pole placement (PP) and an SMC controller. The first two are simulated in MATLAB and the last two are employed in the STTW robot.

1) GESO: The GESO is designed as

$$\begin{aligned}\dot{\hat{\boldsymbol{x}}}&=\bar{\boldsymbol{A}}\hat{\boldsymbol{x}}+\bar{\boldsymbol{b}}_u u+\boldsymbol{L}(\boldsymbol{y}_m-\hat{\boldsymbol{y}}_m)\\\hat{\boldsymbol{y}}_m&=\bar{\boldsymbol{C}}_m\hat{\boldsymbol{x}}\end{aligned}\tag{42}$$

such that $\bar{\boldsymbol{A}}-\boldsymbol{L}\bar{\boldsymbol{C}}_m$ is a Hurwitz matrix. This observer is consistent with Assumption 3 and the gain $\boldsymbol{L}$ are selected such that the poles of the closed-loop system are $[-30,-30,-30,-30]$.

2) H_∞ Controller: Selecting state feedback $u=\boldsymbol{K}_H\boldsymbol{x}$ as the controller architecture, the transfer function from the disturbance d to the controlled output y_o is represented by

$$T(s)=\boldsymbol{C}_o(s\boldsymbol{I}-(\boldsymbol{A}+\boldsymbol{B}_u\boldsymbol{K}_H))^{-1}\boldsymbol{B}_d.$$

Based on structured H_∞ synthesis [31], the control gain is computed by optimizing the H_∞ norm of $T(s)$, such that

$$\|W_l(s)T(s)W_r(s)\|_\infty\leq 1\tag{43}$$

where $W_l(s)=\frac{1}{e_{\max}(1+1/\omega_c s)}$ and $W_r(s)=d_{\max}\frac{1}{1+1/\omega_c s}$, approximately representing that the tracking error should be lower than $e_{\max}$ when the magnitude of constant disturbance is lower

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than $d_{\max}$. In the simulation, we choose $e_{\max} = 1°, d_{\max} = 0.16\ \mathrm{rad/s^2}$ and $\omega_c = 3\ \mathrm{rad/s}$.

3) MPC: The MPC is designed as follows:

$$\begin{aligned} \min_{u_0,\ldots,u_{N-1}} \quad & \sum_{k=0}^{N-1} (\|\boldsymbol{x}_k\|_Q^2 + \|u_k\|_R^2) \\ \text{s.t.} \quad & \boldsymbol{x}_{k+1} = \boldsymbol{A}\boldsymbol{x}_k + \boldsymbol{B}_u \boldsymbol{u}_k, k = 0, \ldots, N \\ & \boldsymbol{x}_0 = \hat{\boldsymbol{x}}_0 \end{aligned} \tag{44}$$

In the simulation, we choose $Q = \mathrm{diag}([5\ 0\ 1]), R = 0.1$.

4) PP controller: The PP controller is designed as

$$u = \boldsymbol{K}\boldsymbol{x} \tag{45}$$

such that $\boldsymbol{A} + \boldsymbol{B}_u\boldsymbol{K}$ is Hurwitz. The gain $\boldsymbol{K}$ is selected such that the poles of the closed-loop system are $[-6, -6, -6]$.

5) SMC: Since the STTW robot is a nonminimum phase system that cannot be transformed to a chain of integrators, the method in [8] is not applicable. The output redefinition method in [14] is applied for the STTW robot. The redefined output is designed as $y_o' = [0\ \ -a_2/a_1\ \ 0]\boldsymbol{x}$ such that $\lim_{t\to\infty} y_o' = \lim_{t\to\infty} y_{o,1}$. Then the normal form of the system can be derived as

$$\begin{aligned} \dot{\boldsymbol{\xi}} &= \begin{bmatrix} 0 & 1 \\ a_2 & a_1/b_1 \end{bmatrix} \boldsymbol{\xi} + \begin{bmatrix} 0 \\ -a_2 \end{bmatrix} \eta + \begin{bmatrix} 0 \\ -a_2 b_1/a_1 \end{bmatrix} u \\ \dot{\eta} &= -\frac{a_1}{b_1}\eta + \begin{bmatrix} \frac{a_1}{b_1} & \frac{a_1^2}{b_1^2 a_2} \end{bmatrix} \boldsymbol{\xi} \end{aligned} \tag{46}$$

where $\boldsymbol{\xi} = [y_o' \quad \dot{y}_o']^T$ and $\eta = \delta - b_1^{-1}\dot{\varphi}$ are the external state and the internal state, respectively. Since $a_1/b_1 > 0$ is determined by the physical model, the zero dynamics is asymptotically stable. Therefore, we only need to stabilize the external dynamics, for which the sliding surface is defined as $s = \dot{y}_o' + \alpha y_o'$, and the control law is designed as

$$u = \frac{a_1}{b_1 a_2}\left(a_2 y_o' + \left(\frac{a_1}{b_1} + \alpha\right)\dot{y}_o' - a_2\eta + \beta \mathrm{sign}(s)\right) \tag{47}$$

where $\alpha > 0,\ \beta > 0$ are tuning parameters. In the experiments, we choose $\alpha = 2.3,\ \beta = 6$ and replace the sign function with the following saturation function to mitigate the chattering issue:

$$\mathrm{sat}(s) = \begin{cases} \mathrm{sign}(s), & |s| \geq \epsilon \\ s/\epsilon, & -\epsilon < s < \epsilon \end{cases}, \qquad \epsilon = 1. \tag{48}$$

D. Numerical Simulation

Comparative simulations regarding the first two controllers and their corresponding ECBCs are conducted in MATLAB Simulink. The simulation results are illustrated in Fig. 3, where "DistFree-Conv" represents "Disturbance-free system under the conventional controller," "Dist-Conv" represents "system with disturbances under the conventional controller" and "Dist-ECBC" represents "system with disturbances under the ECBC." The reference steering angle steps from 0° to 10° at 3 s. The comparisons between "Dist-Conv" and "Dist-ECBC" demonstrate that the ECBCs can effectively eliminate the impact of mismatched disturbance while conventional methods fail to attain accurate tracking. Moreover, the comparisons between "DistFree-Conv" and "Dist-ECBC" over the duration of 3–5 s validate that the ECBCs exhibit equivalent dynamic performances to their corresponding conventional controllers (as stated in Remark 6). Consequently, the desired dynamic performances of an ECBC can be guaranteed by optimizing the parameters of its nominal controller.

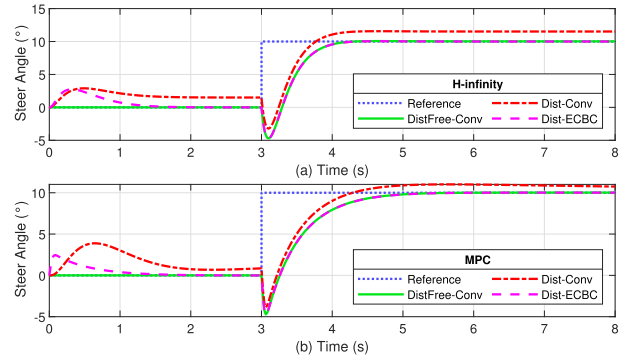

Fig. 3. Comparisons between conventional controllers and ECBC controllers.

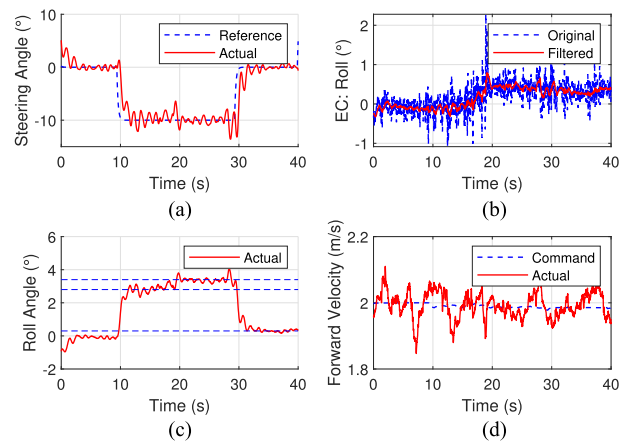

Fig. 4. Results of tracking reference steering angle with ECBC-PP.

E. Experimental Results

The experiments of the STTW robot are made by applying four controllers: ECBC+PP, ECBC+SMC, conventional PP and conventional SMC. Due to the restricted area of the test environment, the reference steering signal is provided by the experimenter with a 2.4 GHz remote transmitter, and thus the reference signals are not completely consistent. In addition, there are several disturbances during the experiments, including 1) varying forward velocity, 2) unmodeled dynamics, 3) sensor error of IMU, 4) gusts and uneven terrains.

Several key variables during an experiment of ECBC-PP are illustrated in Fig. 4. The subfigure (a) shows the accurate tracking of steering angle under ECBC. The subfigure (b) depicts the equilibrium (roll angle) estimated by EC, where the original value is noisy since there are additional high frequency

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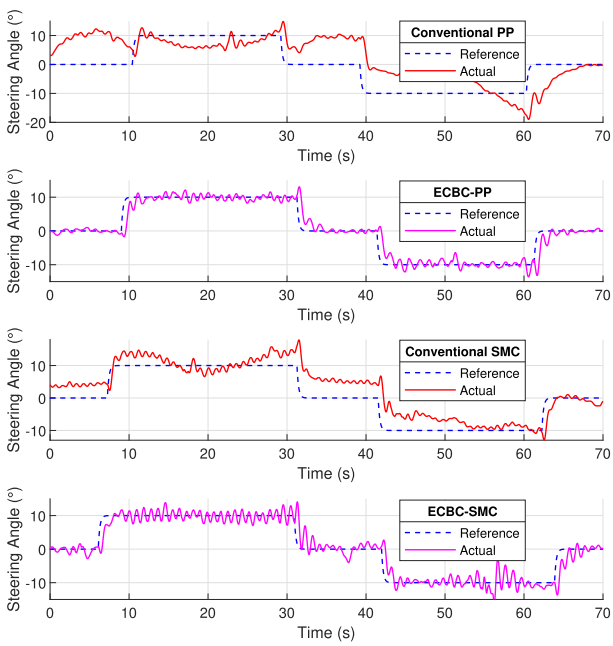

Fig. 5. Comparison between the results of the four controllers.

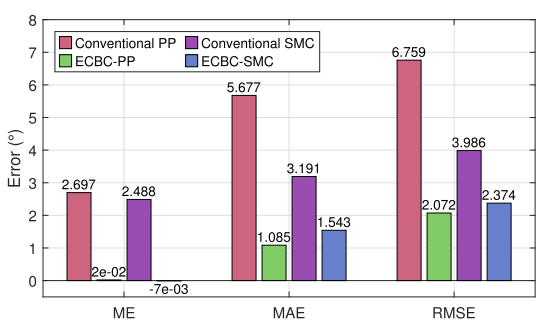

Fig. 6. ME, MAE and RMSE of the tracking errors under the four controllers.

disturbances in the practical system, for instance, the varying forward velocity shown in subfigure (d). To apply the ECBC, a low-pass filter with a cutoff frequency of 3 $\mathrm{rad/s}$ is adopted and the filtered value is drawn with solid curve in subfigure (b). The subfigure (c) shows the roll angle can adaptively adjust the steady value to counteract the slow-varying disturbances and thus guarantee the accurate tracking of output.

The comparison between the experimental results under the four controllers are shown in Fig. 5. The steering angle under ECBC-PP and ECBC-SMC can rapidly track the reference signals without steady-state error, while conventional PP and conventional SMC yield steady-state error over 9° and 5°, respectively. Fig. 6 provides statistical measures of tracking errors for the four controllers, including ME (mean error), MAE (mean absolute error) and RMSE (root mean square error). Remarkably, EC-PP and EC-SMC exhibit almost negligible mean errors of 0.02° and -0.007°, respectively, indicating exceptional tracking accuracy. In contrast, the MEs of conventional PP and conventional SMC are 2.697° and 2.488°, respectively. Furthermore, the EC-PP demonstrates an approximately 80% reduction in MAE and a 70% reduction in RMSE compared to conventional PP. Similarly, EC-SMC exhibits approximately 50% reductions in both MAE and RMSE. These comparisons verify the disturbance rejection and accurate tracking capabilities of the proposed ECBC scheme.

VII. Conclusion

Focusing on the disturbance rejection of linear systems with mismatched disturbances and sensor errors, this paper proposed a new ECBC scheme, which is an extension to existing control methods. The ECBC scheme enables decoupling the design procedures of control methods and disturbance rejection techniques, while retaining nominal performance. Besides, several decoupled criteria were provided for analyzing the observability and the compensability of a perturbed system. The examples showed that the ECBC scheme can be easily applied to a nominal control method, and the experimental results validate its capability of disturbance compensation. In the future, the ECBC scheme will be extended to nonlinear systems with more general disturbances.

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